

Final Project 8.2

Ka-band Satcom Dipole Array with Backing Reflector

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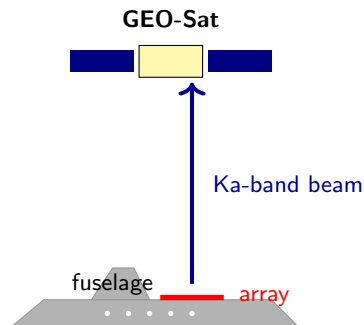
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Why Ka-band Satcom-on-the-move

- In-flight broadband relies on phased arrays conformal to the fuselage skin
- Ka-band Satcom transmit window: **27.5–31 GHz** (3.5 GHz, $\sim 12\%$ fractional)
- Dipoles + PEC backing $\Rightarrow$ low profile, single radiating layer, broadband enough for the band
- Aircraft moves $\Rightarrow$ array must *steer* to the satellite (electronic scan)

Spec: $|\Gamma_{\text{act}}| < -10$ dB across the band, scan electronically, HPBW = 2° .

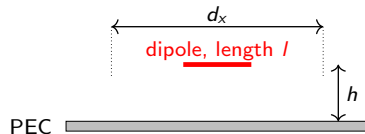


Hard specifications

- Band: $f \in [27.5, 31]$ GHz, $f_c = 29.25$ GHz, $\lambda_c = 10.26$ mm
- $|\Gamma_{\text{act}}|_{\text{dB}} = 20 \log_{10} \left| \frac{Z_{\text{in,act}} - Z_L}{Z_{\text{in,act}} + Z_L} \right| < -10$ dB
- HPBW = 2°
- No grating lobes, no scan blindness

Free design parameters

- Dipole length l , width w
- Reflector height h
- Lattice periods d_x, d_y
- Feed reference Z_L



- 1 **Reuse** the periodic MoM machinery I built in Assignments 5–6: sinusoidal entire-domain basis, Galerkin projection, Floquet-mode sum.
- 2 **Replace** the free-space spectral Green's function with the one that includes the PEC image: only mathematical change, but it controls the entire bandwidth and matching behaviour.
- 3 **Sweep** the design space (h, l, Z_L) on a coarse grid and pick the best in-band $|\Gamma_{\text{act}}|$ at broadside.
- 4 **Verify** scan behaviour: $|\Gamma_{\text{act}}|$ vs θ_0 in E, H, D planes; grating-lobe diagrams at every band corner.
- 5 **Size** the array from the closed-form HPBW $\approx 0.886\lambda/L$ and verify with a windowed pattern simulation.

Galerkin + Floquet reduce the EFIE to a discrete sum:

$$Z_{\text{in,act}}(\theta_0, \phi_0) = -\frac{1}{d_x d_y} \sum_{m_x, m_y} \tilde{G}_{xx}^{ej}(k_{xm}, k_{ym}) [I(k_{xm})]^2 [J_t(k_{ym})]^2$$

- $k_{xm} = k_0 \sin \theta_0 \cos \phi_0 - \frac{2\pi m_x}{d_x}$
- $\tilde{G}_{xx}^{ej} = -\zeta_0(k_0^2 - k_x^2)/(2k_0 k_z)$
- Single sinusoidal basis $\Rightarrow I(k_x), J_t(k_y)$
- Truncation $|m| \leq 25$ converged within 1% on $\text{Im}\{Z\}$

Decomposition

$$Z_{\text{in,act}} = \underbrace{Z_{00}}_{\text{radiates}} + \underbrace{Z_{\text{ho}}}_{\text{evanescent reactance}}$$

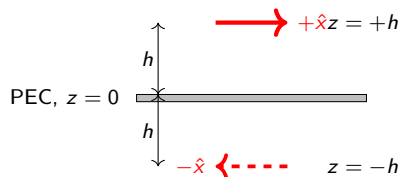
- Z_{00} from the propagating fundamental mode
- Z_{ho} from inter-cell mutual coupling

The one new ingredient: image-theorem Green's function

Image principle. Horizontal $\hat{x}$ -current at $z = +h$ above PEC has an image at $z = -h$ with *opposite* sign.

$$\tilde{G}_{xx,\text{eff}}^{ej}(k_x, k_y; h) = \tilde{G}_{xx,\text{FS}}^{ej}(k_x, k_y) (1 - e^{-j2k_z h})$$

- Propagating modes:
 $|1 - e^{-j2k_z h}|^2 = 4 \sin^2(k_z h)$
- Evanescent: $1 - e^{2|k_z| h} \rightarrow$ **decays** fast in $|k_z| h$
- Quarter-wave broadside: $k_z h = \pi/2 \Rightarrow$ factor
 $= 2$ ($\text{Re}\{Z_{00}\}$ doubles)



One factor changes — it controls the bandwidth.

Grating-lobe-free condition (E-plane, principal scan):

$$\sin \theta_{\text{GL}} = \frac{\lambda}{d_x} - 1 > 1 \quad \Leftrightarrow \quad d_x < \frac{\lambda_{\min}}{2}$$

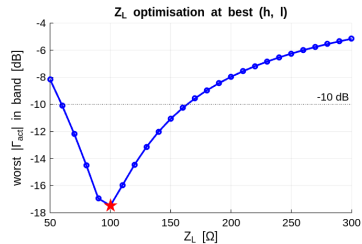
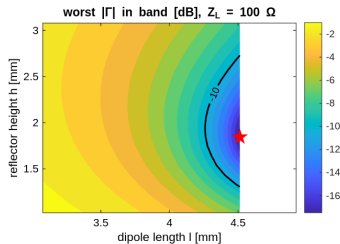
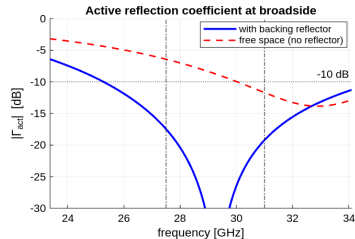
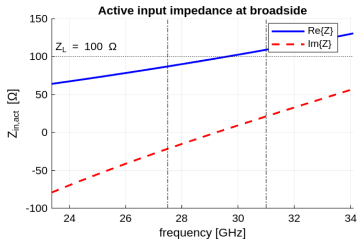
With $f_{\max} = 31$ GHz, $\lambda_{\min} = 9.67$ mm, set $d_x = d_y = \boxed{4.8 \text{ mm}}$.

f [GHz]	λ [mm]	λ/d_x	GL-free?
27.5	10.91	2.273	yes
29.25	10.26	2.137	yes
31.0	9.67	2.014	yes

Result: no first-order Floquet circle ever overlaps the visible region $\Rightarrow$ *no grating lobes, no scan blindness, at any scan angle, anywhere in the band.*

Q1 — Design sweep at broadside

Q1: Ka-band Satcom dipole array — design at broadside ($f = 27.50\text{--}31.00$ GHz)
 $h = 1.85$ mm, $l = 4.51$ mm, $w = 0.40$ mm, $d_x = d_y = 4.80$ mm, $Z_L = 100\ \Omega$



Q1 — The reflector is what makes the bandwidth

Optimum unit cell:

- $h = 1.85 \text{ mm} = 0.180 \lambda_c$ (sub-quarter-wave!)
- $l = 4.51 \text{ mm} = 0.440 \lambda_c$
- $w = 0.40 \text{ mm}$
- $d_x = d_y = 4.80 \text{ mm} = 0.467 \lambda_c$
- $Z_L = 100 \Omega$

Worst-case in-band $|\Gamma_{\text{act}}|$ at broadside:

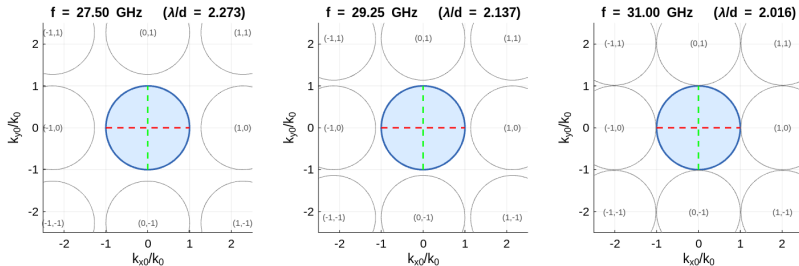
- with backing reflector: -17.5 dB
- free space (no reflector): -3 dB (*out of spec*)

Two key sub-textbook choices

- $h \approx \lambda_c/5.5$, *not* $\lambda_c/4$:
short reflector flattens the resonance,
broadens the band
- $l = 0.95 d_x$:
longest dipole that fits the cell;
pushes resonance just above f_c

Q2 — No grating lobes anywhere in the band

Q2: grating-lobe diagram for the chosen lattice ($d = 4.8$ mm)

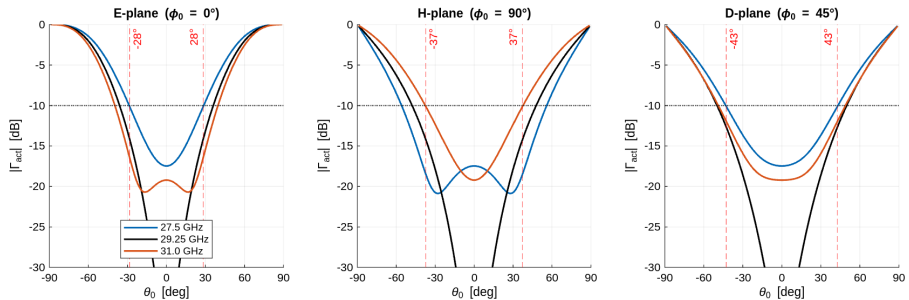


Visible region (blue) never overlaps any higher-order Floquet circle (grey).
First-order Floquet centres at $(\lambda m_x/d_x, \lambda m_y/d_y)$ are always more than 1 unit away.

Q2 — $|\Gamma_{\text{act}}|$ vs scan angle, three planes, three frequencies

Q2: active reflection coefficient vs scan angle, three frequencies in the band

$h = 1.85$ mm, $l = 4.51$ mm, $w = 0.40$ mm, $d_x = d_y = 4.80$ mm, $Z_L = 100 \Omega$



Q2 — Why is the E-plane the bottleneck?

Fundamental Floquet term of $Z_{\text{in,act}}$ on the principal scan trajectory:

$$Z_{00}(\theta_0) = \frac{\zeta_0}{2 d_x d_y} \frac{k_0^2 - k_{x0}^2}{k_0 k_{z0}} \left(1 - e^{-j2k_{z0}h}\right) [I(k_{x0})]^2$$

H-plane ($k_{x0} = 0$, $k_{y0} = k_0 \sin \theta_0$)

$$Z_{00} \propto \frac{1}{\cos \theta_0} (1 - e^{-j2k_0 \cos \theta_0 h})$$

$1/\cos \theta_0$ up, reflector factor stays ~ 2
 $\Rightarrow \text{Re}\{Z\}$ rises smoothly, easy to keep matched.

E-plane ($k_{x0} = k_0 \sin \theta_0$, $k_{y0} = 0$)

$$Z_{00} \propto \cos \theta_0 (1 - e^{-j2k_0 \cos \theta_0 h}) [I]^2$$

$\cos \theta_0$ down and $I(k_{x0})$ rolls off with scan
 $\Rightarrow \text{Re}\{Z\}$ drops fast, mismatch hits -10 dB first.

Result: system scan limit $\theta_{0,\text{max}} \approx \pm 28^\circ$ across the full band (E-plane, f_{hi}).

Q3 — How many elements for $\text{HPBW} = 2^\circ$?

Closed form:

uniform $N \times N$, broadside:

$$\text{HPBW} \approx \frac{0.886 \lambda}{N d} \text{ [rad]}$$

For $\text{HPBW} = 2^\circ$, $d = 4.8 \text{ mm}$:

f [GHz]	required N
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27.5	58
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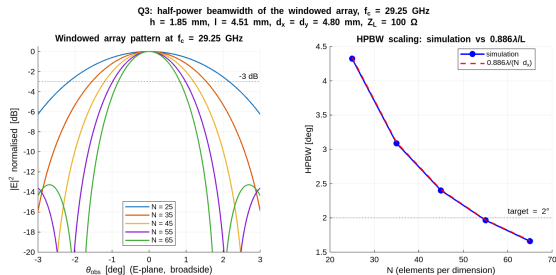
29.25	55
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31.0	52
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Recommendation: $55 \times 55 = 3025$

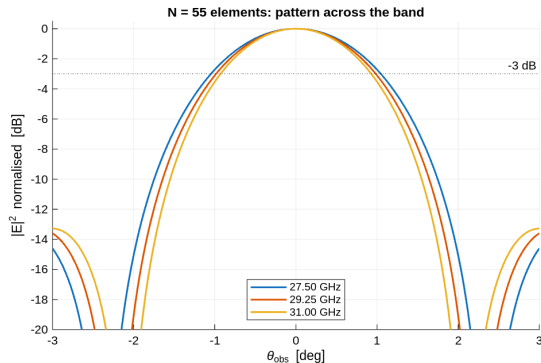
elements

(or 58×58 for strict worst-case)



Simulation overlaps the closed-form $0.886\lambda/L$ exactly.

Q3 — Beamwidth across the band



Pattern of the chosen 55×55 array.

f [GHz]	HPBW [deg]
27.5	2.09
29.25	1.97
31.0	1.85

- HPBW shrinks at higher f
- 55×55 hits the spec at f_c , 4% overshoot at f_{lo}
- 58×58 guarantees the worst case

- ➊ **Backing reflector via image theorem** ($\tilde{G}_{xx,\text{eff}} = \tilde{G}_{xx,\text{FS}} [1 - e^{-j2k_z h}]$): the only mathematical change vs Assignments 5–6, drives the entire bandwidth and matching behaviour.
- ➋ **Designed unit cell**: $h=1.85$ mm, $l=4.51$ mm, $w=0.4$ mm, $d_x=d_y=4.8$ mm, $Z_L=100\ \Omega$. Worst-case broadside $|\Gamma_{\text{act}}| = -17.5$ dB (7.5 dB below spec).
- ➌ **Sub-half-wave lattice** guarantees no grating lobes and no scan blindness anywhere in the band.
- ➍ **System scan limit**: $\theta_{0,\text{max}} \approx \pm 28^\circ$ (E-plane at f_{hi}); H-plane $\pm 37^\circ$, D-plane $\pm 43^\circ$.
- ➎ **Array sizing**: $55 \times 55 = 3025$ elements deliver HPBW = 2° at f_c . **58×58 = 3364 elements** for strict worst-case compliance.

Thank you — questions?